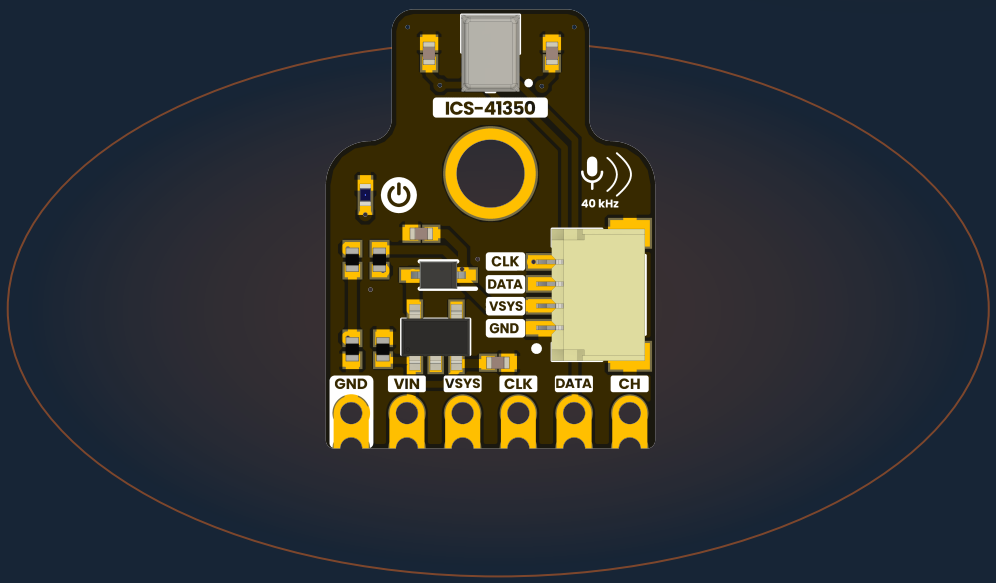
PRODUCT REFERENCE MANUAL

DEVLAB PDM AUDIO MODULE

DevLab PDM ICS-41350 MEMS Microphone

MPN: UE0148

Version 0.1.0



The module combines the multi-mode ICS-41350 MEMS microphone, PDM channel selection, and a fixed 3.3 V regulator.

PDM AUDIO | MULTI-MODE | CHANNEL SELECT | MEMS

DOCUMENT MAP

Contents

Description

Applications
Hardware Features

1 The Module

1.1 Board Identification
1.2 Interface Hardware

2 Electrical Characteristics

2.1 ICS-41350 Characteristics
2.2 Electrical Precautions

3 Functional Overview

3.1 Block Diagram
3.2 Microphone and PDM Signal Path
3.3 Clock Modes and Control Signals
3.4 Power Section
3.5 Communication Interface
3.6 Library Features
3.7 Board Topology
3.8 Electrical Schematic

4 Connectors and Pinout

4.1 Pinout Diagram
4.2 Edge Connections
4.3 Four-Position Connector
4.4 Solder Options

5 Board Operation

5.1 Wiring and First Power-Up
5.2 Arduino Library Example
5.3 Other Capture Examples and Checks

6 Mechanical Information

6.1 Top View
6.2 Bottom View
6.3 Board Dimensions

7 Company Information

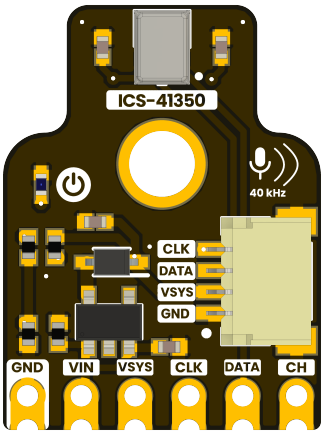
8 Reference Documentation

9 Appendix

9.1 Technical Basis
9.2 Document Scope
9.3 Document Control
9.4 Board Schematic

Description

The UNIT DevLab PDM ICS-41350 MEMS Microphone is a compact digital-audio module built around the bottom-port TDK InvenSense ICS-41350. The microphone contains a MEMS element, impedance-converter amplifier, and fourth-order sigma-delta modulator that produces a 1-bit pulse-density-modulated output.



Applications

- Voice capture and recognition prototypes
- Microphone-array development
- Camera, security, and surveillance audio
- Low-power ambient sound analysis and keyword spotting
- Ultrasonic sensing experiments in high-performance mode

Hardware Features

- ICS-41350 bottom-port digital MEMS microphone
- PDM clock and data interface
- Clock-selected low-power, standard, high-performance, and sleep modes
- Sensor channel selection for shared-data stereo configurations
- AP2112K-3.3TRG1 fixed 3.3 V regulator
- Six 2.54 mm edge/header connections
- Four-position, 1.0 mm-pitch right-angle connector
- Orange indicator LED and two solder options shown on the V1.1 artwork

1 The Module

1.1 Board Identification

Item	Value
Product	DevLab PDM ICS-41350 MEMS Microphone
Manufacturer part number	UE0148
Hardware revision shown	V1.1
Microphone	TDK InvenSense ICS-41350
Digital audio interface	PDM
Product-reference revision	0.1.0

1.2 Interface Hardware

The module provides six 2.54 mm edge connections and a four-position, 1.0 mm-pitch right-angle connector. The connector carries CLK, DATA, VSYS, and GND. Its signal order follows the board pinout in Section 4.

2 Electrical Characteristics

2.1 ICS-41350 Characteristics

The electrical table specifies ICS-41350 sensor characteristics. The module power circuit includes a fixed 3.3 V regulator.

Parameter	Low-power mode	Standard mode	High-performance mode	Scope
Clock frequency	400–800 kHz	1.0–3.3 MHz	4.1–4.8 MHz	Sensor CLK
Sensitivity, typical	−26 dB FS	−26 dB FS	−32 dB FS	Sensor, 1 kHz and 94 dB SPL
SNR, typical	63 dBA	64 dBA	64 dBA	Sensor
Supply current, typical	185 µA	430 µA	650 µA	Sensor at VDD = 1.8 V, no load
Acoustic overload point, typical	120 dB SPL	120 dB SPL	126 dB SPL	Sensor, 10% THD

Parameter	Minimum	Typical	Maximum	Scope
Sensor supply voltage, VDD	1.65 V	—	3.63 V	ICS-41350 supply pin
Sleep-mode clock	—	—	Below 200 kHz	ICS-41350 CLK
Sleep current	—	12 µA	20 µA	ICS-41350
Digital input high, VIH	0.65 × VDD	—	—	ICS-41350 digital input
Digital input low, VIL	—	—	0.35 × VDD	ICS-41350 digital input
Regulated output	—	3.3 V nominal	—	AP2112K-3.3 regulator

The sensor VDD range applies at the microphone supply pin. The module routes VIN through its regulator and Schottky diode to VSYS, which supplies the microphone. The schematic defines the module power path; the sensor VDD range does not specify a module VIN input range.

2.2 Electrical Precautions

- The host and module share a common ground during operation.
- The microphone drives DATA toward the host.
- The host applies CLK and CH within the powered microphone's logic domain.
- The bottom acoustic port remains clear of debris, flux, adhesive, and enclosure obstructions.
- The board schematic shows the VIN to VSYS supply path and the J2 power connection.

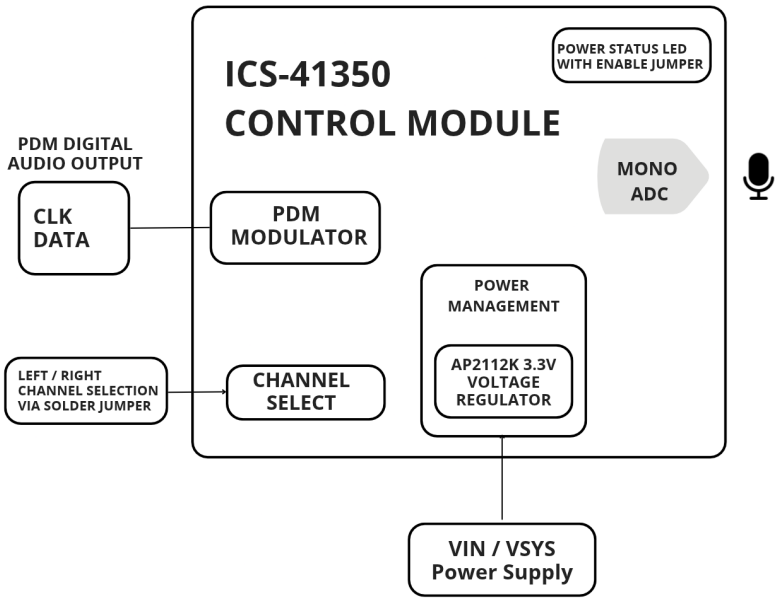
3 Functional Overview

The DevLab PDM ICS-41350 MEMS Microphone combines the TDK InvenSense ICS-41350 digital microphone with a 3.3 V regulator and a channel-select circuit. The microphone has no registers: the host supplies CLK, the microphone returns a 1-bit pulse-density-modulated (PDM) stream on DATA, and the CLK frequency alone selects the sensor mode.

The module does not produce audio samples by itself. A host with a PDM receiver decimates and filters the bitstream into PCM audio.

3.1 Block Diagram

The block diagram shows the main functional blocks of the UE0148 module and the signals available.



UE0148 PDM ICS-41350 microphone module block diagram.

The host drives CLK; the microphone drives DATA toward the host. VIN feeds a 3.3 V regulator whose output reaches the microphone through the VSYS rail.

3.2 Microphone and PDM Signal Path

Sound enters the ICS-41350 (MK1) through its bottom acoustic port. A MEMS sensing element and impedance-converter amplifier feed a fourth-order sigma-delta modulator, and the DATA pin carries its 1-bit PDM output. The host supplies CLK, then decimates and filters the PDM bitstream to obtain audio samples. The ICS-41350 datasheet describes this path in its functional block diagram and “Theory of Operation. ”

The microphone has no register map and no configuration bus. Only three signals reach it from the connectors: CLK and DATA for the audio stream, and CH for channel selection. The VSYS rail powers it.

3.3 Clock Modes and Control Signals

The sensor has no mode register. Its clock frequency selects the mode:

Sensor mode	CLK frequency	Typical sensor current	Typical sensitivity
Sleep	Below 200 kHz	12 µA	No data output
Low power	400–800 kHz	185 µA	−26 dB FS
Standard	1.0–3.3 MHz	430 µA	−26 dB FS
High performance	4.1–4.8 MHz	650 µA	−32 dB FS

The table lists **sensor** characteristics from DS-000047. Clock ranges between listed modes lie outside the specified operating modes. Below 200 kHz, DATA becomes high impedance. After a valid clock starts, the microphone takes up to 20 ms to wake and settle. High-performance mode supports ultrasonic capture up to 40 kHz only when the host decimation filter and acoustic path also preserve that band.

Channel selection. The ICS-41350 SELECT pin low selects the right channel (DATA1); high at sensor VDD selects the left channel (DATA2). Right-channel bits are clocked on the rising edge and left-channel bits on the falling edge. The microphone releases DATA outside its selected half-cycle, so one left and one right microphone can share a data line and a clock. The datasheet specifies a shared PDM data line without a pull-up or pull-down.

The UE0148 schematic routes board CH to the sensor SELECT net. R5 (0 Ω) ties that net to a node that R3 (10 kΩ) pulls to ground, and a solder option can connect the same node to VSYS. The circuit therefore selects the **right** channel by default, and the assembled solder option determines the effective selection. The host PDM receiver selects the same channel as the microphone. The four-position connector carries CLK and DATA, but not CH; channel selection through that connector uses the board option.

3.4 Power Section

The schematic identifies U3 as an AP2112K-3.3 fixed regulator supplied from VIN. Its output passes through the Schottky diode D2 to VSYS, and the microphone VDD pin is on VSYS. The schematic annotates VIN as input only and VSYS as bidirectional (I/O). Both rails are available on the edge connections; J2 exposes VSYS.

The regulator uses 1 µF capacitors C8 (input) and C9 (VSYS). The microphone VDD pin has a 100 nF capacitor (C6) and a 200 pF capacitor (C1) to ground. An orange power LED (D1) connects from VSYS through R1 (4.7 kΩ) and a solder option (LED ON ENABLE) to ground.

The ICS-41350 VDD operating range of 1.65–3.63 V applies at the sensor pin; the module supply path determines the voltage at that pin. The module VIN input range, current consumption, and backfeed behavior through VSYS require board validation and do not follow from the regulator or diode component ratings.

3.5 Communication Interface

The ICS-41350 provides a single digital interface, PDM, which connects directly to microcontrollers and audio processors that include a PDM receiver.

PDM Interface

- 1-bit pulse-density-modulated output, host-supplied clock
- Sensor clock range from 400 kHz to 4.8 MHz in the specified operating modes
- Two selectable channels (right or left) through the SELECT pin

- Shared DATA line for a left/right microphone pair

The host drives CLK and receives DATA. The sensor does not process audio into PCM; the host receiver converts the PDM stream to PCM, using dedicated hardware (for example, the PDM-to-PCM converter of the ESP32-S3) or software filtering. The audio sample rate follows from the PDM clock frequency and the host decimation ratio, and the PDM clock frequency also selects the sensor mode.

3.6 Library Features

The DevLab_AudioInput library for Arduino-compatible boards captures mono audio from PDM and I²S microphones and returns signed 16-bit PCM samples.

PDM capture depends on the PDM receiver of the host board. For example, the ESP32-S3 includes a hardware PDM-to-PCM converter, while the ESP32-C6 does not. The following table summarizes the functions used with the UE0148.

Function	Library call	Description
Initialization	`beginPDM(clkPin, dataPin, sampleRate, slot)`	Starts PDM capture on the selected pins and returns `true` on success. Returns `false` if another source is already active (call `end()` first), the board has no PDM-to-PCM converter, or initialization fails. `slot` selects the receiver channel (`I2S_PDM_SLOT_LEFT` or `I2S_PDM_SLOT_RIGHT`).
Data acquisition	`read(buffer, samples, timeoutMs)`	Places mono signed 16-bit PCM samples in `buffer` and returns the number of samples read; returns 0 when no data arrives within the timeout.
Gain	`setGainPercent(percent)`, `getGainPercent()`	Scales PCM samples by 50 % to 800 %; values outside this range are limited to it. 100 % leaves the samples unchanged, and the scaled result is clipped to the signed 16-bit range.
Source status	`source()`	Reports whether the active input is PDM, I²S, or none.
Shutdown	`end()`	Stops capture so another source can start.

The receiver slot is passed to `beginPDM()` and is not detected automatically. It must match the microphone channel: the UE0148 default (CH low) uses `I2S_PDM_SLOT_RIGHT`, while the library default is `I2S_PDM_SLOT_LEFT`. The PDM clock is generated by the host peripheral, so measuring CLK with a logic analyzer shows the sensor mode in use.

Default values applied by the library

Parameter	Default value
Sample rate	48 kHz
Receiver slot	Left (`I2S_PDM_SLOT_LEFT`)
Sample format	Signed 16-bit PCM, mono
Software gain	100 %
`read()` timeout	10 ms

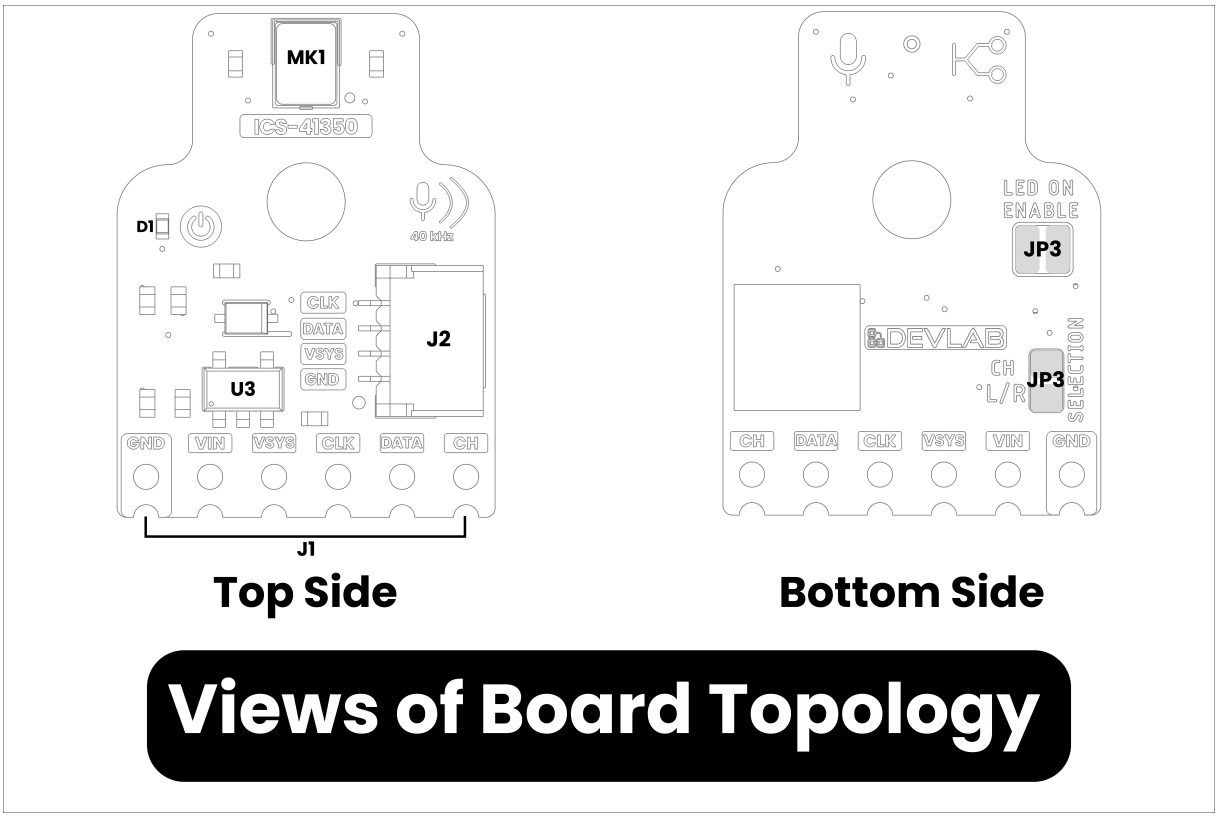
Use `setGainPercent()` to change the gain after `beginPDM()`.

Example sketches

Sketch	Source	Output
`AudioInputBasicRead`	UE0148 repository	Absolute peak of each PCM16 block as text, 115200 baud
`LegacySerialStream`	UE0148 repository	Binary PCM16 stream through the ESP32 I²S PDM driver, 2000000 baud
`BasicRead`	Library `examples/PDM`	Peak level as text, 115200 baud
`SerialPlotter`	Library `examples/PDM`	Decimated samples as text, 115200 baud
`SerialSamples`	Library `examples/PDM`	Every PCM16 sample as text, 8 kHz, 115200 baud
`SerialStream`	Library `examples/PDM`	PCM16 mono, 48 kHz, 2000000 baud
`ComputerAudio`	Library `examples/PDM`	PCM16 mono, 16 kHz, 921600 baud, played by `pc_audio.py`

3.7 Board Topology

The released topology drawing locates the microphone (MK1), regulator (U3), indicator LED (D1), edge connections (J1), four-position connector (J2), and solder options on the top and bottom sides of the board.



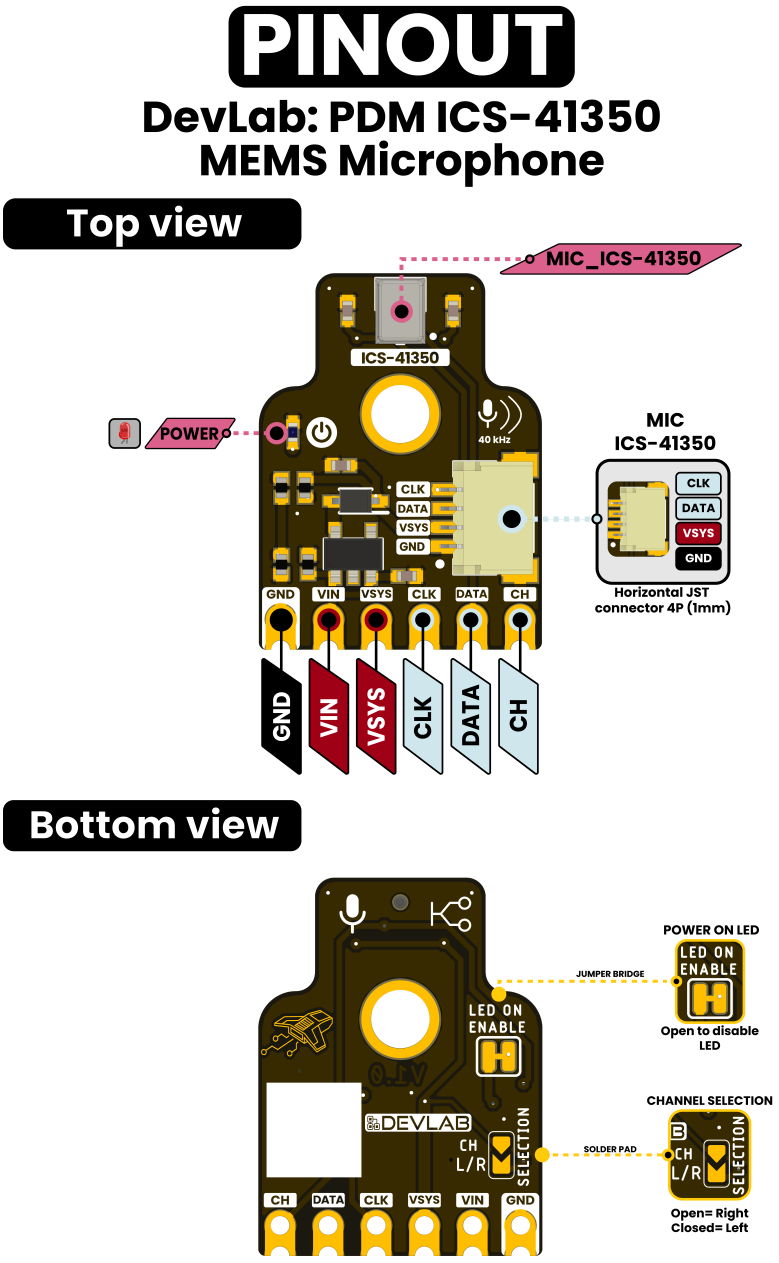
UE0148 PDM ICS-41350 top and bottom board topology.

3.8 Electrical Schematic

The complete V1.1 schematic, covering the microphone, power supply, connector, channel selection, and indicator circuits, is available in the original schematic PDF, which allows inspecting small component and net labels at full resolution.

4 Connectors and Pinout

4.1 Pinout Diagram



4.2 Edge Connections

Board label	Direction	Function
GND	Power	Common ground
VIN	Input	Regulator input
VSYS	Power	System rail supplying the microphone
CLK	Input	PDM clock input
DATA	Output	PDM microphone data

Board label	Direction	Function
CH	Input	Connects to ICS-41350 SELECT through R5; low selects right, high selects left

4.3 Four-Position Connector

The V1.1 top artwork labels the connector signals, from top to bottom in the rendered view, as CLK, DATA, VSYS, and GND. J2 is a four-position, 1.0 mm-pitch right-angle connector. The board pinout shows signal order by physical position.

4.4 Solder Options

The bottom artwork identifies CH SELECTION L/R and LED ON ENABLE options. R3 pulls channel selection low, so the default channel is right. The assembled solder bridges determine channel selection and LED operation.

5 Board Operation

This example uses the DevLab_AudioInput library to convert PDM input to signed 16-bit mono PCM. The example is shown on an ESP32-S3; other hosts with a PDM receiver can use the same connections, adjusting the pins and sketch settings to the board.

5.1 Wiring and First Power-Up

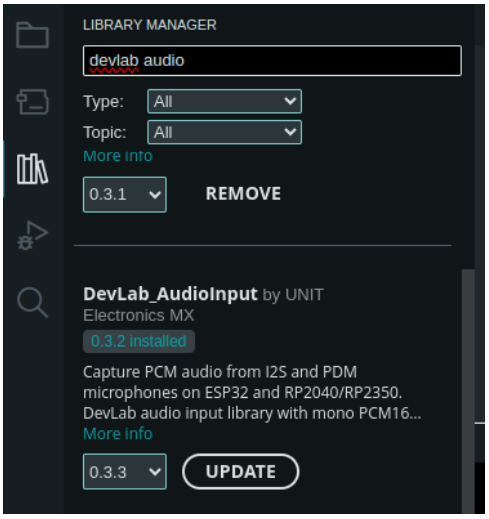
The connection uses a clear acoustic port, a common ground, and a supply that follows the module power circuit. The board revision and solder options determine the effective channel selection. The sensor VDD range describes the microphone pin rather than the module `VIN` input.

UE0148 signal	Example ESP32-S3 pin	Direction
`CLK`	GPIO2	Host PDM clock output
`DATA`	GPIO3	Host PDM data input
`CH`	Board default low	Right channel
`GND`	GND	Common ground
`VIN` or `VSYS`	Supply matching the board power circuit	Board power

The UE0148 circuit pulls `CH` low and selects the right PDM channel by default. If the assembled board selects left, the sketch uses `I2S_PDM_SLOT_LEFT`. The GPIO numbers identify host pins; the four-position connector carries `CLK`, `DATA`, `VSYS`, and `GND`.

5.2 Arduino Library Example

The example requires the library in the Arduino sketchbook and an ESP32-S3 with Arduino-ESP32 3.x. Arduino Library Manager lists `DevLab_AudioInput`:



DevLab_AudioInput entry in Arduino Library Manager.

The complete AudioInputBasicRead sketch reports the absolute peak of each captured PCM16 block at 115200 baud:

```
#include <DevLab_AudioInput.h>

#if defined(ARDUINO_ARCH_RP2040)
#define PDM_CLK_PIN  10
#define PDM_DATA_PIN 11
#else
#define PDM_CLK_PIN  GPIO_NUM_2
#define PDM_DATA_PIN GPIO_NUM_3
#endif

DevLab_AudioInput pdm;
int16_t samples[512];

void setup() {
  Serial.begin(115200);
  delay(1000);

  if (!pdm.beginPDM(PDM_CLK_PIN, PDM_DATA_PIN)) {
    Serial.println("PDM initialization failed.");
    while (true) delay(1000);
  }
}

void loop() {
  size_t n = pdm.read(samples, 512);
  if (n == 0) return;

  // Peak level of the block
  int32_t peak = 0;
  for (size_t i = 0; i < n; i++) {
    int32_t v = abs(static_cast<int32_t>(samples[i]));
    if (v > peak) peak = v;
  }
  Serial.println(peak);
}
```

The Serial Monitor displays the peak at 115200 baud. The peak changes with sound level and represents digital PCM amplitude. A logic analyzer shows the actual PDM clock frequency and its corresponding sensor mode.

5.3 Other Capture Examples and Checks

The library's `examples/PDM` folder also provides `SerialPlotter`, `SerialStream`, `SerialSamples`, and `ComputerAudio`. Its stock examples use the library's default left slot; the UE0148 default uses `I2S_PDM_SLOT_RIGHT`. The existing LegacySerialStream sketch uses the ESP-IDF driver directly and selects the board's drawn right slot. It is retained as an experimental legacy sketch.

The library's PDM sketches use CLK on GPIO2 and DATA on GPIO3 on an ESP32-S3, the same pins as the wiring table in Section 5.1. To use them with the UE0148 default channel, pass `I2S_PDM_SLOT_RIGHT` as the fourth argument of `beginPDM()`.

Serial streaming protocol. `SerialStream` and `ComputerAudio` send raw audio over the serial port instead of text:

Item	SerialStream	ComputerAudio
Sample rate	48 kHz	16 kHz
Baud rate	2000000	921600

Item	SerialStream	ComputerAudio
Format	PCM16, mono, little-endian, no delimiters between samples	PCM16, mono, little-endian, no delimiters between samples
Start	Sends a RAW line, then streams	Waits for R, answers with a RAW line, then streams
Commands	R starts or resynchronizes, S stops, G<percent> sets gain (for example G200 for 200 %)	R starts, S stops
Gain	100 % until changed with G	Fixed at 800 % in the sketch

Close the Arduino Serial Monitor before connecting another application to the serial port. The receiver uses the sample rate and baud rate in the table. `SerialSamples` prints every PCM16 sample as text at 8 kHz for inspecting the waveform; at 115200 baud it can lose samples in real time, so use `SerialStream` or `ComputerAudio` to capture audio.

Playing audio on a computer. `ComputerAudio` has a companion Python script, `pc_audio.py`, in the same folder. It plays the stream through the computer's audio output:

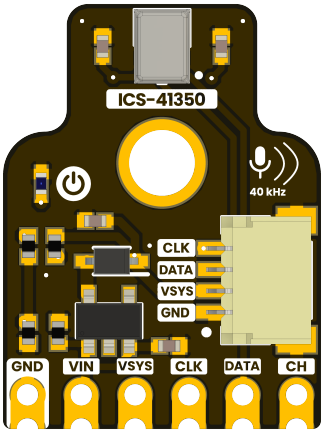
1. Install the dependencies with `pip install pyserial sounddevice`.
2. Upload the `ComputerAudio` sketch to the host board.
3. Close the Arduino Serial Monitor.
4. From the folder that contains `pc_audio.py`, run the script with Python, passing the serial port of your host board as the argument.

The script sends R until it receives the RAW header and then plays the stream until you press Ctrl+C. The `SAMPLE_RATE` and `BAUD_RATE` values in the sketch and in `pc_audio.py` must match. The optional `--device` argument selects another audio output device; the `sounddevice` module can list the devices available on your computer.

If initialization fails, the host configuration checks PDM-to-PCM support, the Arduino core version, and GPIO availability. If captured samples remain near zero, the signal check covers power, clock, DATA, and the receiver slot relative to CH. If samples clip or show noise, the check covers the supply, clock waveform, acoustic port, and host gain.

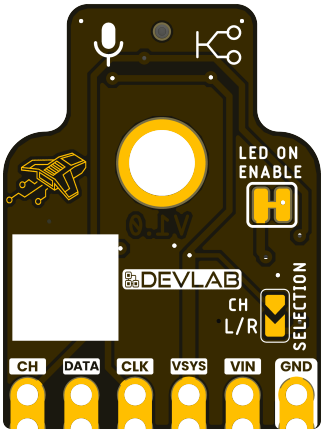
6 Mechanical Information

6.1 Top View



UE0148 PDM microphone top view.

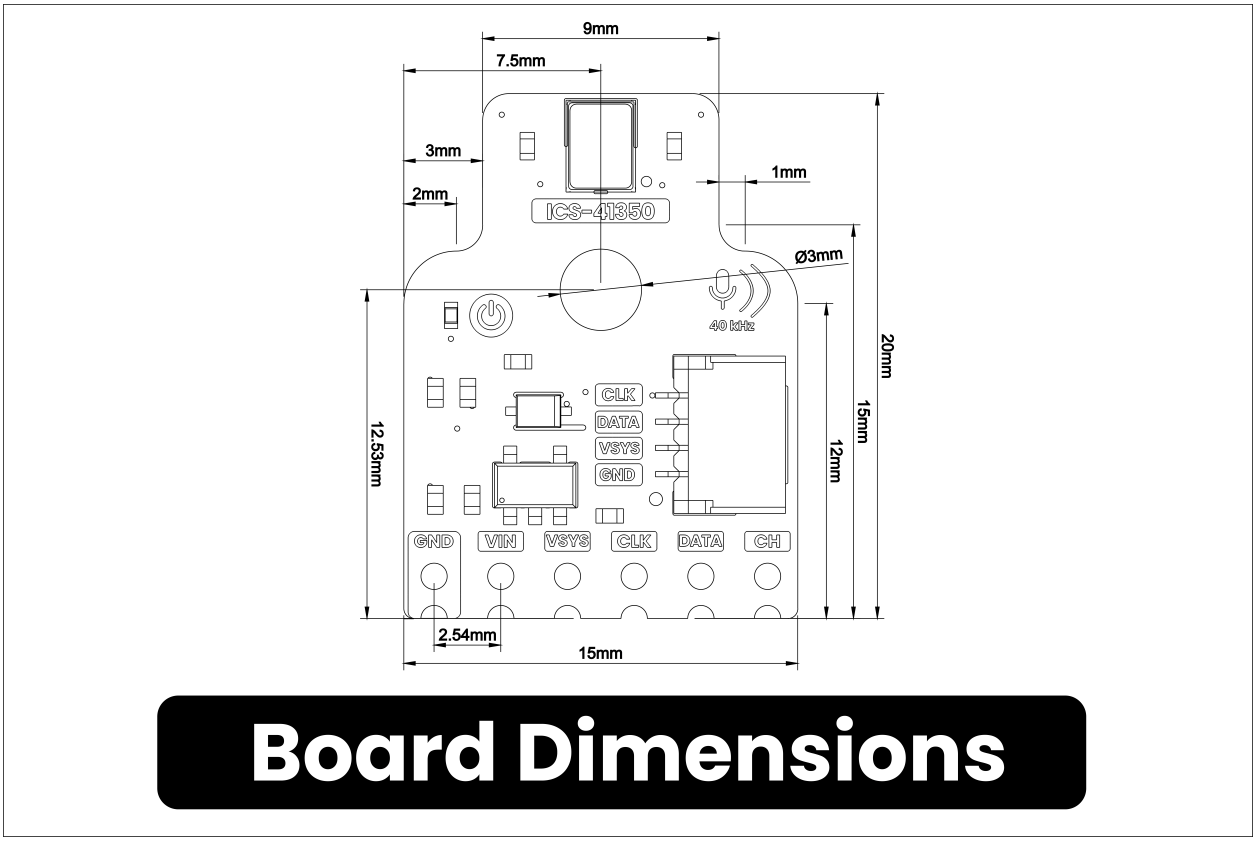
6.2 Bottom View



UE0148 PDM microphone bottom view.

The bottom acoustic opening remains clear of adhesive, debris, and enclosure features.

6.3 Board Dimensions



UE0148 PDM microphone dimension drawing.

The labeled drawing provides the board dimensions for mechanical layout. The physical assembly determines connector clearance and manufacturing tolerances.

7 Company Information

Item	Information
Company	UNIT Electronics
Website	https://uelectronics.com/
Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico

The repository in Section 8 provides the current product documentation.

8 Reference Documentation

Reference	Location
Repository	UNIT-Electronics-MX/ unit_devlab_pdm_ics_41350_mems_microphone
Hardware notes	Hardware README
Pinout PDFs	English · Spanish
Pinout images	English PNG · Spanish PNG
Microphone datasheet	TDK InvenSense ICS-41350
Local sensor datasheet	DS-000047 rev. 1.1
Module schematic	UE0148 schematic
Capture library	DevLab_AudioInput
Module block diagram	UE0148 block diagram
Library-based example	UE0148 AudioInputBasicRead
Top view	3D board view
Bottom view	3D board view
CAD model	STEP

The local microphone source is TDK InvenSense document DS-000047, revision 1.1, dated 2016-06-23. The sensor characteristics in this reference follow that datasheet. The module connections follow the UE0148 schematic and pinout, and the visible labels follow the V1.1 board artwork.

9 Appendix

9.1 Technical Basis

This product reference follows the V1.1 board artwork for visible signal labels, the UE0148 schematic for circuit connections, and TDK InvenSense datasheet DS-000047 revision 1.1 for ICS-41350 characteristics.

9.2 Document Scope

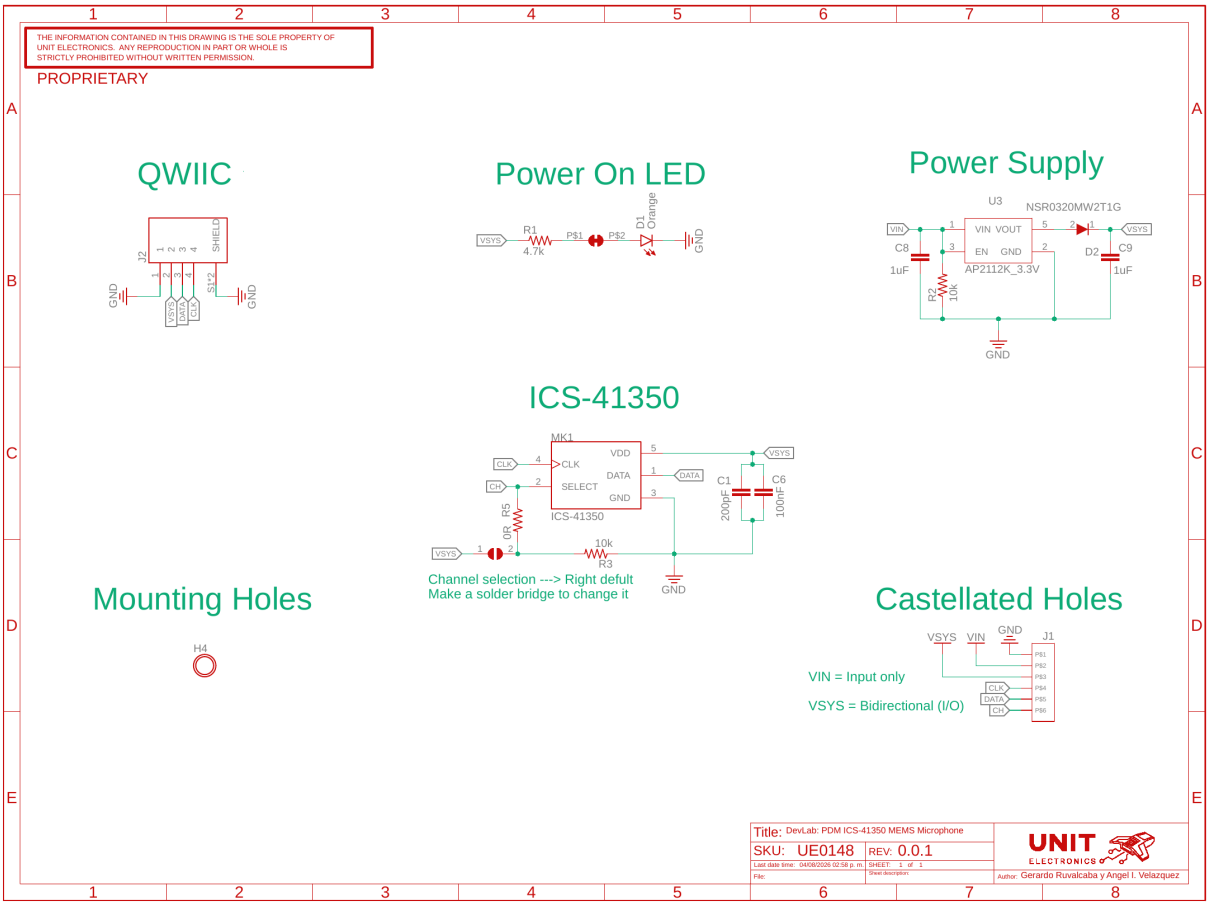
- Section 2 lists the ICS-41350 sensor characteristics and the module's nominal regulated voltage.
- Sections 3 and 4 describe the signal path, power circuit, and board connections.
- Section 5 shows an ESP32-S3 capture example using the DevLab_AudioInput library.
- Section 6 shows the board views and dimension drawing.
- Section 9.4 presents the complete UE0148 board schematic.

9.3 Document Control

Field	Value
Product	DevLab PDM ICS-41350 MEMS Microphone
Manufacturer part number	UE0148
Hardware artwork covered	V1.1
Product-reference revision	0.1.0
Publication date	2026-08-03
Firmware example	ESP32-S3 PDM capture in Section 5

9.4 Board Schematic

The UE0148 schematic shows the microphone, power supply, connector, channel selection, and indicator circuits.



UE0148 PDM ICS-41350 board schematic.